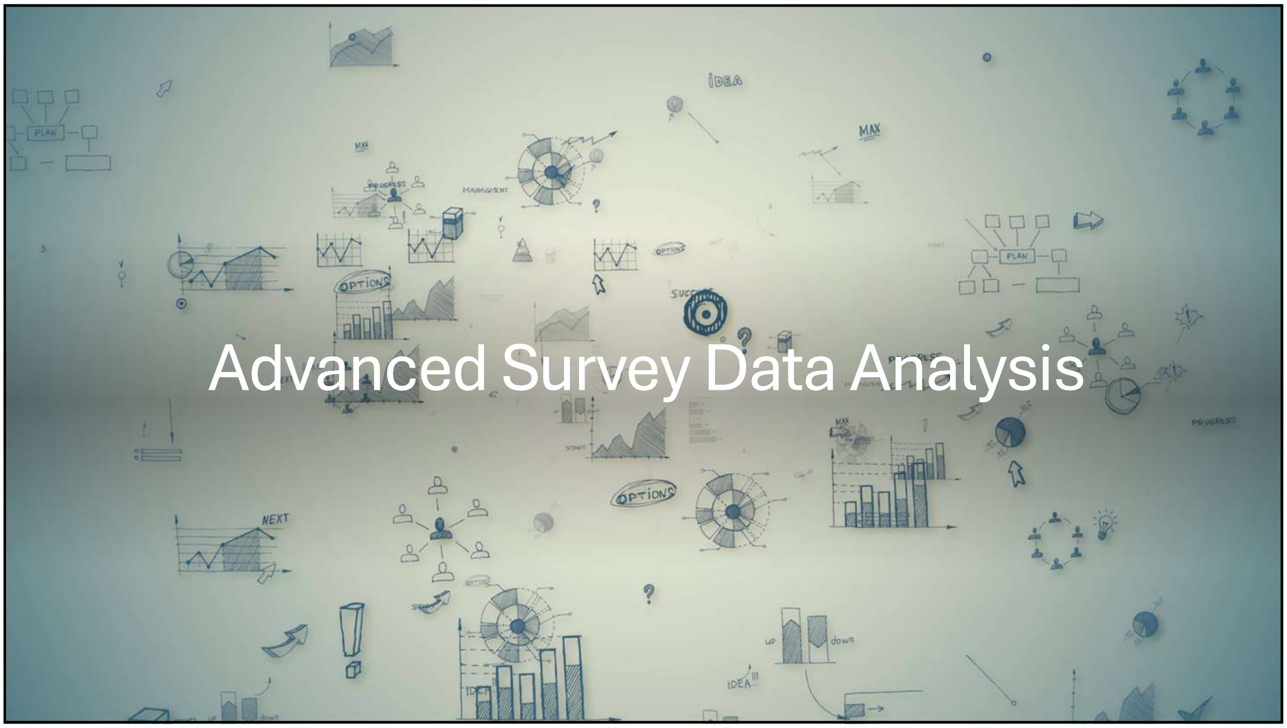

Advanced Survey Data Analysis

78

What is Unstructured Text Data in Surveys?

- **Structured questions**: Likert scales, multiple choice, ratings
- **Unstructured questions**: Open-ended feedback such as:
  - “What did you like most?”
  - “What would you improve?”

79

Why use them?

- Because they capture **the voice of the respondent**
- What they felt, noticed, or needed, in **their own words**.

80

Challenges in Working with Text Responses

- **Messy spelling and grammar**
  - "instractor was gr8" → you know what they meant, but the algorithm doesn't.
- **Short, incomplete responses**
  - "Good." but good in what way? Content? Teacher? Pacing?
- **Ambiguity and context**
  - "Too fast" — is that a good thing or a complaint?
- These issues make **automation and quantification** difficult but not impossible.

81

NLP Concepts

- Token: A single word (or unit)
- Document: One complete response
- Corpus: A collection of responses (all survey answers)

82

Text Cleaning and Preprocessing

- Tokenization: process of **splitting text into individual units**, usually words or punctuation marks.
- **Example:**
  - **Raw Text**:
    “The instructor was AMAZING and very clear!!”
- **Tokenized:**
    ["The", "instructor", "was", "AMAZING", "and", "very", "clear", "!!"]
  - Break sentences into **tokens** to work with each word individually.

83

Text Cleaning and Preprocessing

- **Lowercasing and Removing Punctuation:**
  - Tokenized, usually **normalize** text to remove irrelevant differences:
  - Convert everything to **lowercase**
  - Remove **punctuation** that doesn't add meaning
- **Example:**
  - Tokenized:
    ["The", "instructor", "was", "AMAZING", "and", "very", "clear", "!!"]
- After normalization:
  - ["instructor", "amazing", "clear"]

84

Text Cleaning and Preprocessing

- **Stopword Removal**
  - the, is, and, in, of, to, was
  - These are **syntactic glue** they don't carry significant meaning in most analyses.
- Removing stopwords **reduces noise**, especially when building word frequencies or topic models.

85

Text Cleaning and Preprocessing

- **Stemming :**
  - Removes suffixes
  - Fast, but often **inaccurate**
  - "teaching" → "teach“
- **Lemmatization (Smarter):**
  - Uses vocabulary and grammar rules
  - Converts to the **dictionary base form**
  - "better" → "good“
  - "studies" → "study"
- For survey analysis, **lemmatization is preferred.**

86

Sentiment Analysis in Survey Data

- Sentiment analysis, also known as **opinion mining**, is the process of using **NLP** to detect the **emotional tone** behind a piece of text.

87

Why Sentiment Useful in Surveys?

- Don’t need to read hundreds of open-ended responses.
- Enables you to track trends over time
- Helps segment customer/student groups by satisfaction
- Can be combined with structured data for deeper insights

88

Limitations and Ethical Considerations

- Sentiment analysis **does not understand** text but makes **pattern-based guesses**.
- Mislabeling can lead to **wrong conclusions**.
- Short texts, emojis, sarcasm, and multilingual input often **break** traditional tools.
- Always verify automated insights with **human review** for critical decisions.

89

Text Frequencies and Visualization

- Sometimes, simple word counts give you the clearest insights.
- Especially in survey analysis, **just knowing what words are mentioned most** can tell us:
  - What people care about
  - What topics keep recurring
  - Where issues are concentrated

90

Bag of Words (BoW)

- BoW is a **simple model** that:
  - Ignores grammar, word order, and context
  - Treats each document as a “bag” of words
  - Just counts **how often** each word appears
- It doesn’t care **where** or **how** a word appears — only **how many times**.

91

Bag of Words (BoW)

- Survey Question: “What would you improve in this course?”
- Responses:
  - “More feedback on assignments.”
  - “Too slow sometimes.”
  - “Better materials and less repetition.”

Word	Frequency
feedback	1
assignments	1
slow	1
materials	1
repetition	1

92